

# Reward-Channel Separability in Multimodal Chart RLVR: A Cross-Rollout Bonus and a Negative-Sample Caveat

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## Abstract

Chart question answering requires a model to parse the visual structure, extract the relevant data, and reason to a final answer. Whether these components respond independently to different training signals, however, is not well understood. We study this with two opposing interventions: Hierarchical Correct-Path Consistency (HCPC), a cross-rollout bonus rewarding consistent extraction and diverse reasoning among correct responses, and Negative Sample Reinforcement (NSR; ?), which removes gradient signal from correct responses entirely.

We evaluate on ChartQA (in-distribution) and the held-out ChartFC benchmark, which is excluded from training and serves as our out-of-distribution (OOD) test. NSR collapses format compliance from 96.8% to 31.6% on ChartQA with no gain in answer accuracy over GRPO. Per-tag analysis traces the failure to the structural tags reward training had to teach, pointing to reward-signal starvation as the cause. HCPC achieves the best Pass@4 on ChartQA (0.828 vs. 0.816 for GRPO) but degrades on ChartFC, where reliance on structured extraction becomes a liability. Combining the two, NSR+HCPC recovers GRPO-level accuracy on ChartFC while format compliance barely recovers. The fact that accuracy returns but format does not suggests the two are distinct behavioral channels, and that reward update rules from single-component settings should be re-evaluated before applying them to richer reward landscapes.

## 1 Introduction

Recent chart-reasoning vision-language models are trained via reinforcement learning with verifiable rewards (RLVR), typically using GRPO with multi-component rewards. Answering questions about charts requires two sequential operations that are rarely treated as distinct (????). A model must first extract the underlying data table from

the visual layout, then reason over that table to reach an answer. The first operation has one correct outcome; the second admits many valid paths. Current training pipelines collapse this distinction. The dominant recipe applies GRPO (?) with verifiable rewards across chart-type identification, table reconstruction, process conformity, and answer accuracy (????), reinforcing all correct rollouts uniformly regardless of which operation they got right or how. The practical cost is gradual: the policy over-commits to whichever solution paths dominate the training distribution, eroding the diversity that supports robustness under distribution shift and coverage at higher  $K$  in Pass@ $K$  (?).

We propose Hierarchical Correct-Path Consistency (HCPC), a cross-rollout bonus that treats the two operations differently. Among rollouts that get both the table and the answer right, HCPC rewards consistent table extraction and diverse reasoning traces. Consistency is appropriate at the extraction layer, where there is one correct table and reliable recovery is the goal. Diversity is appropriate at the reasoning layer, where multiple paths to the same answer reflect genuine competence. A ground-truth-anchored filter ensures the bonus targets correct-path quality rather than surface agreement among arbitrary rollouts.

To explicitly isolate what HCPC is and is not doing, we introduce a contrastive diagnostic that applies opposing pressure to the same problem. Negative Sample Reinforcement (NSR; ?) targets the same diversity goal by the opposite route: masking gradient signal on correct rollouts and penalizing only incorrect ones. In single-component math RLVR, this works (?). In chart RLVR, it does not. Porting NSR unchanged collapses format compliance, specifically the structured XML tags and table markup that the pipeline depends on, on both in-distribution and out-of-distribution data, with no corresponding gain in answer accuracy. Per-tag analysis traces the failure to exactly the struc-

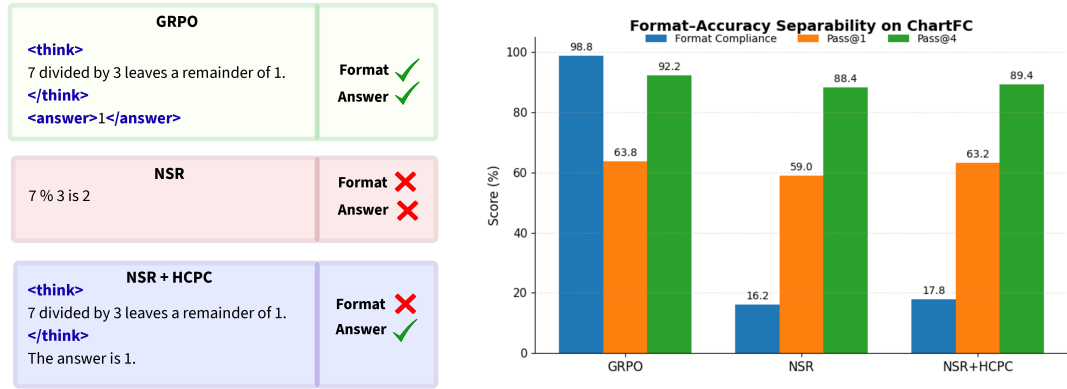


Figure 1: Format and answer accuracy are separable channels in chart RLVR. **Left:** a stylized rollout under each training regime. GRPO emits the required structural tags and the correct answer; NSR drops both; NSR+HCPC recovers the answer while the structured tags remain collapsed. **Right:** ChartFC (OOD) metrics for the same three regimes. Adding HCPC on top of NSR restores Pass@1 and Pass@4 to near GRPO levels, but format compliance barely moves from NSR’s collapsed value (16.2% to 17.8%). Accuracy returns; format does not. The dissociation is direct evidence that these two behaviors respond independently to training interventions.

tural tags that reward training had to teach, pointing to reward-signal starvation as the cause. The result is informative precisely because the failure is clean: something about the multimodal, multi-component reward landscape breaks an assumption that held in the simpler setting.

Adding HCPC to NSR recovers competitive answer accuracy out-of-distribution while format compliance stays near zero. Accuracy returns; format does not. This dissociation is the central finding of the paper. Format compliance and answer accuracy are distinct behavioral channels in chart RLVR, and they can be manipulated independently. The broader implication is not just that NSR fails here, but that reward channels in this setting are separable and should be designed with that separability in mind. A training objective that treats them as one signal will, at best, trade one off against the other; at worst, disrupt both while appearing to recover neither. Our contributions are as follows:

1. **HCPC (§3.2, §5.2).** A cross-rollout bonus applying level-specific pressure to table extraction and reasoning separately. HCPC improves robustness across multi-sample and table-dependence metrics while maintaining parity with the GRPO baseline on single-sample accuracy.
2. **NSR transferability analysis (§5.3).**

Porting NSR unchanged into our multi-component chart-reasoning training pipeline collapses format compliance without accuracy gain. Per-tag analysis identifies reward-signal starvation as the mechanism, localized to structural tags the base model does not emit unprompted.

3. **Separability of format and accuracy (§5.4).** Adding HCPC to NSR recovers out-of-distribution accuracy all the way back to the GRPO baseline, while format compliance barely moves from NSR’s collapsed level. One channel snaps back; the other does not. This is direct evidence that the two are distinct behavioral channels that respond independently to training interventions. We argue reward components in chart RLVR should be designed with this separability explicit.

## 2 Related Work

**Chart reasoning with RLVR.** Most current chart-reasoning VLMs are trained with GRPO and multi-component verifiable rewards. Chart-RVR (?) set the template, combining rewards for format, chart-type, table reconstruction, process conformity, and answer accuracy. Recent work extends this along orthogonal axes: reasoning-trace supervision and CoT distillation (??), richer training data (??), grounding reflection (?), and agen-

tic pipelines that wrap base models rather than modifying training (??). Our work sits next to these. We keep Chart-RVR’s reward backbone unchanged and ask a different question: how the choice of advantage rule and a cross-rollout reward interact with that backbone.

**Advantage-rule variants in RLVR.** GRPO (?) treats correct and incorrect rollouts symmetrically, normalizing rewards within a group. ? split this into positive- and negative-sample reinforcement (PSR and NSR), and show that NSR alone, masking gradients on correct rollouts and penalizing only incorrect ones, matches or beats GRPO on math benchmarks. Related work studies RLVR in adjacent settings (????). All assume a single scalar reward, the setting NSR was designed for. We are the first to port NSR into a multi-component, multimodal reward landscape, and to identify a clean failure mode: structural rewards saturate early, pushing well-formed but answer-wrong rollouts above NSR’s correctness threshold, which masks them and starves the structural side of positive gradient.

**Group-level signals and rollout-set diversity.** Self-consistency (?) is the closest existing cross-rollout signal, but operates at inference, aggregating  $K$  samples by majority vote for calibration, not training. ETTRL (?) preserves diversity at test time via entropy control. HCPC differs in three ways: it acts at training time as a reward bonus; it is anchored to ground-truth correctness, not majority agreement, so consistently wrong groups get no bonus; and it applies different objectives at different levels, consistency for table extraction and diversity for reasoning. We use Pass@ $K$  (?) as our coverage diagnostic; HCPC’s level-specific design targets the higher- $K$  regime where symmetric advantage rules fall behind.

**Format and accuracy as separable behaviors.** The dissociation we find, where format compliance and answer accuracy move in opposite directions under the same intervention, has no direct precedent in chart RLVR we are aware of. The closest adjacent observation is that CoT prompting can degrade VLM accuracy in chart settings (?), suggesting structured reasoning and correctness already trade off through some channel. Our result is stronger than a tradeoff: NSR+HCPC recovers GRPO-level accuracy while format compliance stays at NSR’s collapsed level. This makes reward-

component independence a real design choice for VLM RLVR, not a side note.

### 3 Methodology

We build on Chart-RVR (?) as a reward backbone and introduce two interventions: HCPC (Hierarchical Correct-Path Consistency), a group-level bonus rewarding level-specific structure among correct rollouts, and NSR (?), a contrastive gradient-masking strategy that we port to chart RLVR as a diagnostic.

#### 3.1 Preliminaries: Chart-RVR Reward Backbone

We adopt Chart-RVR’s per-rollout reward structure, which supervises each output component independently. Rollouts are expected in the format `<think><type/> <table/> reasoning</think> <answer/>`. Four components are active:  $R_{\text{fmt}}$  (structural compliance with the required XML template),  $R_{\text{acc}}$  (answer correctness against  $y^*$ ),  $R_{\text{tbl}}$  (table extraction quality assessed by partial structural alignment against  $T^*$ ), and  $R_{\text{type}}$  (exact match of the predicted chart type against the ground-truth type  $c^*$ ). Their sum gives the base reward:

$$R_{\text{base}}(o_i) = R_{\text{fmt}} + R_{\text{acc}} + R_{\text{tbl}} + R_{\text{type}}.$$

Three Chart-RVR components are inactive in our setup. A length reward  $R_{\text{len}}$  (encouraging mid-range rationale lengths) and a token-count reward  $R_{\text{tok}}$  (firing only when each required tag appears exactly once) are both disabled because the base model prepends a native reasoning block that yields duplicate tags and unconstrained length, making these signals noisy on our setup. Chart-RVR also defines a process reward measuring similarity to the ground-truth reasoning chain; we disable it because it conflicts with  $D_{\text{reason}}$ ’s diversity objective.

#### 3.2 HCPC: Hierarchical Correct-Path Consistency

HCPC augments  $R_{\text{base}}$  with a group-level bonus computed exclusively over correct rollouts. The design enforces *level-specific coherence*: consistency where one factually correct answer exists (table extraction) and diversity where multiple paths are valid (reasoning steps). Cross-rollout statistics are conditioned on ground-truth correctness rather

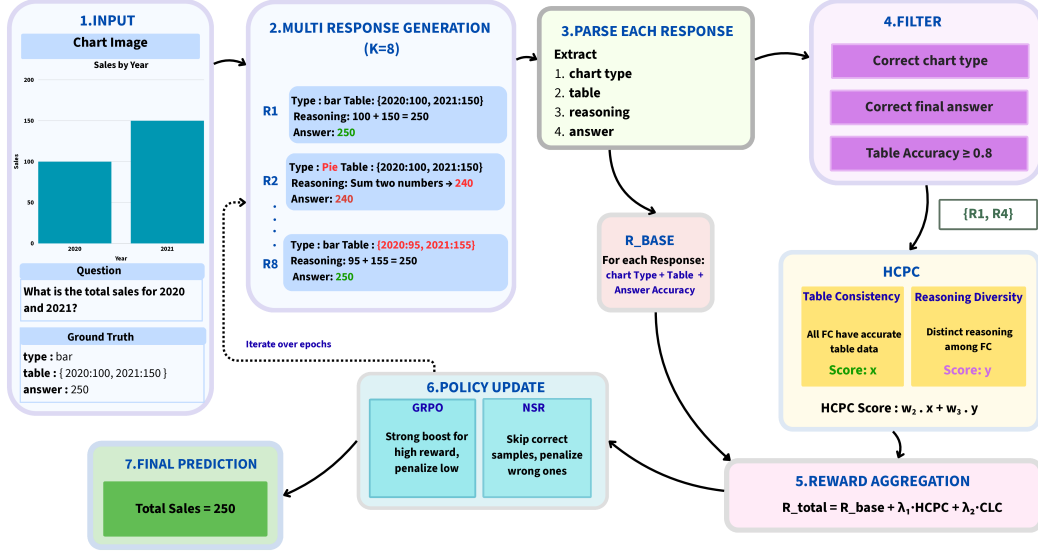


Figure 2: Training pipeline. The multi-component reward supervises each rollout independently through format, table, accuracy, and length rewards. HCPC adds a group-level bonus computed over rollouts that pass the correct-path filter ( $G^+$ ), rewarding consistent table extraction together with diverse reasoning. We compare two advantage estimators over the same per-rollout reward: standard GRPO and NSR, which masks the gradient of correct rollouts.

than majority agreement (?), preventing shared extraction errors from producing a spuriously high consistency signal.

### 3.2.1 Correct-Path Filtering

Let  $G = \{o_1, \dots, o_K\}$  be the rollouts for a prompt with ground truth  $(T^*, y^*)$ . We define the *correct-path subset*  $G^+ \subseteq G$  via three sequential gates.

**Gate 1 (format).** The rollout must be structurally parseable with all required tags and a valid table block.

**Gate 2 (table fidelity).** The extracted table  $\hat{T}_i$  must satisfy  $\text{sim}(\hat{T}_i, T^*) \geq \delta$  with  $\delta=0.8$ , using cosine similarity over MiniLM-L6 embeddings (?) on serialized table strings. This excludes rollouts that reach the correct answer from a misread table.

**Gate 3 (answer).** The predicted answer  $\hat{y}_i$  must match  $y^*$ : relative error  $\leq 5\%$  for numeric answers, normalized substring match for string answers.

All three gates must pass. An answer-correct rollout with a wrong table has not followed a genuine reasoning path; rewarding its cross-rollout properties would propagate extraction errors into the group signal.

### 3.2.2 Level-Specific Group Metrics

On  $G^+$  we compute three metrics targeting distinct trajectory levels.  $C_{\text{type}}$  is the fraction of rollouts in  $G^+$  that predict the modal chart type  $\hat{c}_{\text{mode}}$ .

For table and reasoning levels, let  $\langle f \rangle_{G^+}$  denote the mean of  $f(i, j)$  over all distinct pairs  $i < j$  in  $G^+$ :

$$C_{\text{table}} = \langle \text{sim}(\hat{T}_i, \hat{T}_j) \rangle_{G^+},$$

$$D_{\text{reason}} = 1 - \langle \text{sim}(\hat{w}_i, \hat{w}_j) \rangle_{G^+}.$$

$C_{\text{table}}$  measures pairwise agreement among extracted tables; correct rollouts should converge on the one ground-truth reading.  $D_{\text{reason}}$  is pairwise dissimilarity among reasoning strings  $\hat{w}_i$ ; high values indicate diverse inferential paths to the same answer.  $C_{\text{type}}$  and  $C_{\text{table}}$  enforce convergence where the correct answer is unique;  $D_{\text{reason}}$  enforces divergence where multiple paths are valid. All similarities use cosine distance over MiniLM-L6 embeddings.

### 3.2.3 Reward Integration

The consistency component of the group bonus is shared equally among all rollouts in  $G^+$ :

$$B_{\text{cons}} = \frac{|G^+|}{K} (w_{\text{type}} C_{\text{type}} + w_{\text{table}} C_{\text{table}}).$$

The diversity component is assigned per-rollout via a uniqueness score  $u_k = 1 - \bar{s}_k$ , where  $\bar{s}_k$  is rollout  $k$ 's mean cosine similarity to all other reasonings in  $G^+$ . The per-rollout HCPC reward is

then:

$$R_{\text{HCPC}}(o_i) = \begin{cases} B_{\text{cons}} + w_{\text{reason}} u_k & \text{if } o_i \in G^+, \\ 0 & \text{otherwise,} \end{cases}$$

with  $w_{\text{type}}=1.0$ ,  $w_{\text{table}}=2.0$ ,  $w_{\text{reason}}=1.5$ , and  $R_{\text{HCPC}}=0$  when  $|G^+|<2$ . The total reward is:

$$R_i = R_{\text{base}}(o_i) + R_{\text{HCPC}}(o_i).$$

In practice  $D_{\text{reason}}$  is small ( $\sim 0.05$  at  $K=4$ ), so the dominant signal is  $w_{\text{table}} C_{\text{table}}$ . HCPC is silent when  $|G^+|<2$ ; on ChartQA this occurs 28.8% of the time ( $|G^+|=0$  on 17.2% of groups,  $|G^+|\geq 3$  on 56.6%), exceeding the naïve Bernoulli bound of 15.7% at  $p=0.62$  due to within-group visual correlation.

### 3.3 Diagnostic Baseline: NSR

Negative Sample Reinforcement (?) pursues the same diversity goal as HCPC by the opposite route: it removes gradient from correct rollouts entirely and penalizes only incorrect ones. In single-component math RLVR this improves Pass@ $K$  (?). We port it to chart RLVR unchanged to test whether that finding transfers to a multi-component reward setting.

**Masking rule.** After GRPO normalizes advantages within a group, we zero out the advantage of any rollout with  $R_i \geq \theta$ , where  $\theta = 0.5 R_{\text{max}}$ .  $R_{\text{max}}$  is summed from all defined reward ceilings in the Chart-RVR specification regardless of whether each component is active:  $R_{\text{max}}=11.0$  for the base configuration and  $R_{\text{max}}=15.5$  with HCPC ( $11.0 + w_{\text{type}} + w_{\text{table}} + w_{\text{reason}}$ ), giving  $\theta=5.5$  and  $\theta=7.75$  respectively. Because two base components are inactive (§3.1), the achievable ceiling is lower; the threshold is therefore conservative, masking only the highest-scoring rollouts. The threshold is fixed rather than group-relative, so it does not shift with the per-step distribution.

**NSR+HCPC.** When combined, HCPC bonus inflates  $R_i$  for rollouts in  $G^+$ , making them more likely to be masked by NSR. Rollouts outside  $G^+$  receive no HCPC bonus and are more likely to retain a negative gradient. HCPC membership thus indirectly determines which rollouts drive the policy update under NSR.

## 4 Experimental Setup

### 4.1 Data and Training Setup

Training uses the first 1,000 samples of sanchit97/chart-rvr-grpo-train (?), ChartRVR’s CoT training mixture drawn from

ChartQA (?), PlotQA, and ChartFC (?). Samples are selected sequentially by index. ChartFC samples in the training mixture come from the training split of ?; the OOD evaluation probe (§4.2) uses the disjoint test split. PlotQA samples are included in training for domain coverage but excluded from evaluation, which targets chart QA and fact-checking directly.

We fine-tune Qwen2.5-VL-3B-Instruct with LoRA ( $r=8$ ,  $\alpha=16$  on  $W_q, W_v$ ) using AdamW at learning rate  $1 \times 10^{-6}$ , effective batch size 4 (per-device 1, gradient accumulation 4),  $K=4$  rollouts per prompt at temperature 0.8 and top- $p=1.0$ , over 2 epochs. No KL penalty is applied ( $\beta_{\text{KL}}=0$ ). We evaluate five configurations: BASE (untuned Qwen2.5-VL-3B-Instruct), GRPO, GRPO+HCPC, NSR, and NSR+HCPC. GRPO uses the standard group-normalized advantage  $\hat{A}_i = (R_i - \bar{R}) / \max(1, s_R)$ ; NSR additionally zeros out advantages on rollouts with  $R_i \geq \theta$  (§3.3).

### 4.2 Evaluation Protocol

Evaluation samples 4 generations per test prompt at temperature 0.8 and top- $p=0.95$  on a 500-sample test slice of each benchmark. **ChartQA** (?) is in-distribution; **ChartFC** (?) is held out as the OOD probe, differing from ChartQA on task format (fact-checking vs. open-ended QA) and visual style.

We report Pass@1, unbiased Pass@4 (?), and format compliance Fmt%. Two additional metrics are central to this paper.

**Per-tag emission rate.** For each required tag ( $\langle \text{think} \rangle$ ,  $\langle \text{answer} \rangle$ ,  $\langle \text{type} \rangle$ ,  $\langle \text{table} \rangle$ , plus proper order), the fraction of rollouts emitting it. This localizes format collapse to specific structural components rather than reporting it in aggregate.

### Table-dependence $\Delta_{\text{tbl}}$ .

$$\Delta_{\text{tbl}} = 2P(\hat{y} = y^* \mid \text{tbl parsed}) - P(\hat{y} = y^* \mid \text{no tbl}),$$

computed per sample on rollout 0 then aggregated across the test slice. A higher  $\Delta_{\text{tbl}}$  indicates the model’s answer correctness depends more strongly on emitting a parseable table. The two arms of the conditional can move for different reasons; we unpack this in §5.2.



| Method    | ChartQA (in-distribution, $n=500$ ) |              |                |             | ChartFC (OOD) |              |                |             |
|-----------|-------------------------------------|--------------|----------------|-------------|---------------|--------------|----------------|-------------|
|           | Pass@1                              | Pass@4       | $\Delta_{tbl}$ | Fmt%        | Pass@1        | Pass@4       | $\Delta_{tbl}$ | Fmt%        |
| BASE      | 0.518                               | 0.774        | -0.031         | 29.8        | —             | —            | —              | —           |
| GRPO      | <b>0.630</b>                        | 0.816        | +0.178         | 96.8        | <b>0.638</b>  | <b>0.922</b> | +0.139         | <b>98.8</b> |
| GRPO+HCPC | 0.622                               | <b>0.828</b> | <b>+0.264</b>  | <b>98.2</b> | 0.606         | 0.918        | -0.059         | 66.4        |
| NSR       | 0.592                               | 0.808        | +0.082         | 31.6        | 0.590         | 0.884        | -0.059         | 16.2        |
| NSR+HCPC  | 0.574                               | 0.800        | +0.147         | 43.0        | 0.632         | 0.894        | +0.002         | 17.8        |

Table 1: Main results.  $\Delta_{tbl}$  is the conditional gain in answer correctness from emitting a parseable table; higher means stronger table-dependence. Bold marks column-best per benchmark. Pass@4 uses the unbiased estimator over 4 generations per prompt.

## 5 Results and Analysis

### 5.1 Main Results

Table 1 reports the five configurations on ChartQA and ChartFC. GRPO training substantially improves both answer accuracy and format compliance over the untuned Base model: Pass@1 rises from 0.518 to 0.630 on ChartQA and Fmt% from 29.8 to 96.8. GRPO+HCPC matches GRPO on Pass@1 (0.622 vs. 0.630) while leading on Pass@4 (0.828 vs. 0.816) and  $\Delta_{tbl}$  (+0.264 vs. +0.178), consistent with HCPC’s design goal of improving rollout-set coverage rather than single-shot accuracy. NSR collapses format compliance to 31.6% on ChartQA and 16.2% on ChartFC with no accuracy gain; the per-tag analysis in §5.3 traces this to reward-signal starvation on structural tags. NSR+HCPC recovers OOD accuracy to 0.632 (statistically tied with GRPO at 0.638; §5.4) while format compliance stays at 17.8%, the format/accuracy dissociation that is the central finding of this paper.

### 5.2 GRPO+HCPC: Rollout Robustness Without Accuracy Regression

HCPC is designed to strengthen correct-path quality by rewarding consistent extraction and diverse reasoning. The metrics that should reflect this are rollout-set robustness and Pass@ $K$  at higher  $K$ , not single-shot Pass@1.

Headline Pass@1 on ChartQA is statistically tied: GRPO 0.630 vs. GRPO+HCPC 0.622 (paired bootstrap -0.8 pp, CI [-3.8, +5.4],  $p=0.77$ ,  $n_{boot}=10,000$ ). On every other in-distribution metric, the direction favors HCPC: the fraction of test samples where all 4 rollouts are correct is 40.8% for HCPC vs. 38.6% for GRPO; the average correct rate per group is 0.628 vs. 0.620; Pass@4 is 0.828 vs. 0.816; table-dependence is  $\Delta_{tbl}=+0.264$  vs. +0.178; format compliance is 98.2% vs. 96.8%. No individual

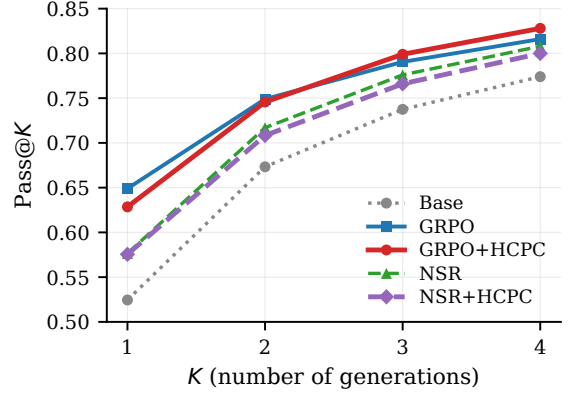


Figure 3: Pass@ $K$  on ChartQA ( $n=500$ ). HCPC’s gain over GRPO grows with  $K$ : tied at  $K=1$ , leading at  $K=3$  and  $K=4$ . The direction is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage at higher  $K$ . Individual differences are not significant at  $n=500$  (Pass@4 boot- $p=0.48$ ).

difference reaches significance at  $n=500$  (Pass@4 boot- $p=0.48$ ), but the joint direction across six metrics in HCPC’s favor is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path quality without trading off single-shot accuracy. Figure 3 shows the Pass@ $K$  curves: HCPC matches GRPO at  $K=1$  and pulls ahead at  $K \geq 3$ , the regime where rollout-set coverage matters.

Unpacking  $\Delta_{tbl}$ :  $P(\hat{y} = y^* \mid tbl)$  is 0.639 for GRPO and 0.632 for HCPC (tied; also tied on the common-table intersection of 462 samples, Appendix C), while  $P(\hat{y} = y^* \mid \text{no tbl})$  is 0.462 for GRPO and 0.368 for HCPC. The gain therefore reflects stronger table-dependence: HCPC drops 9.4 pp more than GRPO when no table is produced, while matching it when a table is produced. The correct-path bonus makes the extracted table load-bearing in the answer pathway.

| Method    | think | answer | type | table | proper order |
|-----------|-------|--------|------|-------|--------------|
| BASE      | 52.2  | 91.2   | 96.6 | 77.2  | 81.6         |
| GRPO      | 97.4  | 97.6   | 100  | 98.6  | 97.4         |
| GRPO+HCPC | 98.8  | 98.2   | 100  | 99.6  | 98.6         |
| NSR       | 57.4  | 94.8   | 96.6 | 72.2  | 85.4         |
| NSR+HCPC  | 57.8  | 91.6   | 94.6 | 83.2  | 84.0         |

Table 2: Per-tag emission rate (%) on ChartQA. Column headers refer to the `<think>`, `<answer>`, `<type>`, `<table>` tags and the “proper order” check. NSR’s collapse is concentrated in the structural tags (think, table) that the base model does not emit unprompted; HCPC under NSR partially recovers table but not think.

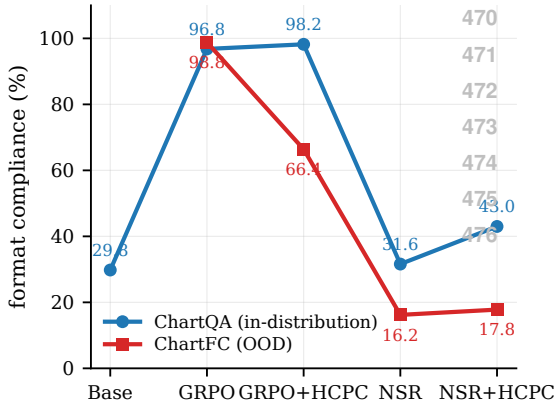


Figure 4: Overall format compliance per method on ChartQA (in-distribution) and ChartFC (OOD). NSR collapses on both; GRPO+HCPC holds ID but degrades OOD; GRPO is the only configuration that keeps format near ceiling under distribution shift.

**GRPO+HCPC OOD limitation.** The same property makes GRPO+HCPC weaker than plain GRPO on the OOD ChartFC probe (Pass@1 0.606 vs. 0.638, Pass@4 0.918 vs. 0.922,  $\Delta_{tbl}$   $-0.059$  vs.  $+0.139$ , format 66.4% vs. 98.8%). The mechanism is plausible from the in-distribution finding: HCPC trains the model to rely on the extracted table, so when table extraction degrades OOD (HCPC’s format compliance drops to 66.4% on ChartFC, indicating the extraction pipeline is less reliable on unfamiliar visual styles), the answer pathway degrades with it. The table-dependence that benefits HCPC in-distribution becomes a liability OOD. This is a genuine limitation of standalone GRPO+HCPC, complementary to the NSR-mitigation result in §5.4; we leave a fix (e.g., a chart-style augmentation in training or a softer GT-anchored filter) to future work.

### 5.3 NSR Breaks VLM Format

Switching the advantage estimator from GRPO to NSR, with no other changes, drops format com-

pliance from 96.8% to 31.6% on ChartQA and from 98.8% to 16.2% on ChartFC. On ChartFC, GRPO leads NSR on Pass@4 by +3.8 pp (CI  $[+0.6, +7.2]$ ,  $p=0.028$ , paired bootstrap; McNemar exact  $p=0.034$ ).

Table 2 localizes the collapse. NSR keeps the tags the base model already emits (`<type>`, `<answer>`, proper order) and loses the ones GRPO has to teach (`<think>`, `<table>`). NSR’s `<think>` rate (57.4%) is essentially the base model’s (52.2%) plus noise; `<table>` rate (72.2%) falls below the base model’s (77.2%).

The mechanism is mechanical.  $R_{fmt}$ , the active structural reward in our setup, saturates quickly: emitting the required tags in the correct order is a learned-once behavior that yields near-maximal contribution from that component. With those rewards saturating, the total per-rollout  $R_i$  for a well-formed rollout reaches the NSR threshold  $\theta = 0.5 R_{max}$  regardless of whether the final answer is correct. NSR therefore masks the advantage of *structure-good, answer-wrong* rollouts, leaving only *structure-bad, answer-wrong* rollouts, which receive negative advantage, to drive the policy. This removes the dominant positive gradient on structural-tag emission, and the policy drifts back toward the base model’s unstructured output. We expect this failure mode to be absent in ?’s math experiments because text-math rewards are dominated by a single answer bit; the implicit “format” (a final `\boxed{}` or a number on its own line) is either trivial or emerges from the answer-correctness signal itself. Verifying this empirically is beyond our current scope.

### 5.4 Format and Accuracy as Separable Channels

On ChartFC, HCPC added to NSR improves over plain NSR on every metric: Pass@1  $0.590 \rightarrow 0.632$  ( $+4.2$  pp), Pass@4  $0.884 \rightarrow 0.894$  ( $+1.0$  pp),  $\Delta_{tbl}$   $-0.059 \rightarrow +0.002$  ( $+0.061$ ), and format  $16.2\% \rightarrow 17.8\%$ . The  $\Delta_{tbl}$  sign flip from negative to zero indicates that table extraction stops actively hurting; the format gain is marginal. Plain NSR is the weakest OOD configuration in Table 1 on every metric, and HCPC closes most of that gap (Figure 5). How the gap is closed (mechanism below) turns out to be more interesting than the gap-closing itself.

**Format/accuracy dissociation.** The way in which HCPC mitigates NSR’s damage is unusual:

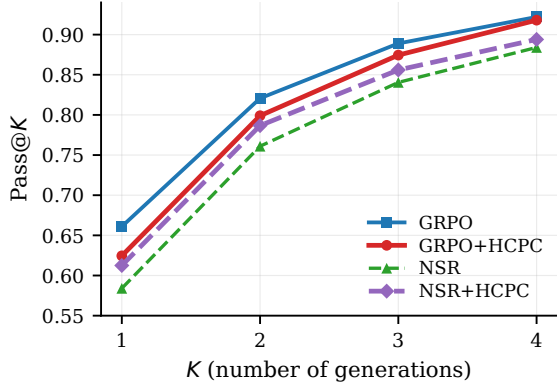


Figure 5: Pass@K on ChartFC (OOD,  $n=500$ ). Plain NSR (green) sits below all other configurations at every  $K$ ; adding HCPC (purple) closes most of the gap to GRPO. BASE was not evaluated on ChartFC.

NSR+HCPC reaches  $\text{Pass}@1 = 0.632$ , statistically indistinguishable from GRPO ( $0.638$ ; paired-bootstrap CI  $[-0.046, +0.060]$ ,  $p=0.85$ ), while format compliance remains collapsed at  $17.8\%$  (vs. GRPO’s  $98.8\%$ ). HCPC under NSR recovers GRPO-level OOD answer accuracy without recovering format. This is the cleanest empirical evidence in the paper that format compliance and answer accuracy are separable downstream behaviors in chart RLVR.

**Why HCPC behaves so differently under GRPO and NSR.** Under GRPO, HCPC behaves as designed: the bonus is added to correct-path rollouts’ rewards, those rollouts get higher GRPO advantage  $(R_i - \bar{R})/s_R$ , and the policy is reinforced toward the correct path. This is the standard positive-gradient interpretation and accounts for the ChartQA results (§5.2).

Under NSR, the same bonus operates through a different channel. HCPC adds its bonus to exactly the correct-path rollouts that NSR then masks out of the gradient, so the bonus does not reach the policy as a direct positive update. But TRL computes the GRPO advantage before the NSR mask is applied, using the group mean  $\bar{R}$  over the full set of rollouts. Adding  $B_{\text{HCPC}}$  to correct rollouts raises  $\bar{R}$ , and for the wrong rollouts that survive the mask this makes their advantage more negative. The correct-path bonus therefore reshapes the reference baseline that the surviving (wrong) rollouts are penalized against, pushing the policy harder away from wrong-path behavior. The mechanism is *baseline shift, not direct reinforcement*. HCPC under NSR strengthens the negative gradient on wrong rollouts without providing any positive gra-

dient on correct ones. This is consistent with what we observe. Answer-level OOD accuracy recovers because wrong-rollout suppression strengthens, while structured-output emission, which requires positive gradient on correct rollouts that neither NSR nor HCPC-under-NSR can provide, remains collapsed.

On ChartQA, adding HCPC on top of NSR also partially restores format ( $31.6\% \rightarrow 43.0\%$  overall, with the recovery concentrated in `<table>`:  $72.2\% \rightarrow 83.2\%$ , Table 2). In-distribution, format partially recovers. Out-of-distribution, accuracy recovers fully while format does not. This asymmetry makes the channel-separability claim concrete.

## 6 Conclusion

We introduced HCPC, a cross-rollout bonus that rewards consistent extraction and diverse reasoning on the correct-path subset of a group. On top of GRPO it produces more robust correct groups and stronger table grounding without sacrificing  $\text{Pass}@1$ . Comparing against NSR, a diversity-preservation recipe imported from text-only math RLVR, we find that porting it unchanged into chart RLVR collapses format compliance: structural rewards saturate fast enough for well-formed rollouts to cross NSR’s threshold regardless of answer quality. HCPC on top of NSR recovers GRPO-level OOD accuracy at near-zero format compliance, providing direct evidence that format and answer accuracy are separable channels of the trained policy. The takeaway for practitioners: advantage-rule asymmetries developed for single-reward source domains can interact destructively with reward components that are new in the target domain. The shape of the multi-component reward mixture deserves first-class consideration when porting RLVR recipes across modalities.

## Limitations

We use a single 3B-parameter backbone (Qwen2.5-VL-3B), a 1,000-sample training subset,  $K=4$  rollouts, and a 500-sample test slice per benchmark. Compute constraints prevented us from training on the full Chart-RVR CoT mixture (34,194 samples), evaluating on the full ChartQA and ChartFC test sets, or running EvoChart (?) and ChartQAPro (?) as style-OOD benchmarks distinct from the joint task-and-style probe ChartFC provides. Scaling to the full



training data, full test sets, and a style-only OOD benchmark is our highest-priority follow-up.

## A Reproducibility

All training and evaluation code, the four trained LoRA adapters, and per-sample evaluation outputs (for the 500-sample test slices of ChartQA and ChartFC, including raw rollout text) are released as supplementary material. Hyperparameters and the exact reward configuration used for every run are listed in the released training configs (configs/experiment.py); the expected output schema is in §B. Pass@ $K$  uses the unbiased estimator of ?. Cross-rollout similarities use MiniLM-L6 embeddings. Paired-bootstrap CIs use  $n_{\text{boot}}=10,000$  resamples; McNemar tests use the two-sided exact binomial form on the discordant pair.

## B Prompt Template and Expected Output Format

The model is prompted to emit responses in a structured XML template. The prompt instructs the model to wrap its reasoning in `<think>...</think>`, identify the chart type, extract the underlying data table, perform step-by-step reasoning, and finally emit the answer in `<answer>...</answer>`. The expected output schema is:

```
<think>
<type>...</type>
<table>{"columns": [...], "rows":
  ↳ [...]}</table>
<step-1>...</step-1>
<step-2>...</step-2>
...
</think>
<answer>...</answer>
```

The per-tag emission rate metric (§4.2) and the format reward  $R_{\text{fmt}}$  are both defined relative to this schema. `<think>` and `<table>` are not natively emitted by Qwen2.5-VL-3B-Instruct without training; the format reward provides the gradient that teaches them.

## C Per-Tag Format Compliance, Both Benchmarks

Table 2 in the main body reports per-tag emission rates for ChartQA. Table 3 extends this to both benchmarks. BASE was not evaluated on ChartFC.

| Method                               | think | answer | type | table | order |
|--------------------------------------|-------|--------|------|-------|-------|
| <i>ChartQA (in-distribution)</i>     |       |        |      |       |       |
| BASE                                 | 52.2  | 91.2   | 96.6 | 77.2  | 81.6  |
| GRPO                                 | 97.4  | 97.6   | 100  | 98.6  | 97.4  |
| GRPO+HCPC                            | 98.8  | 98.2   | 100  | 99.6  | 98.6  |
| NSR                                  | 57.4  | 94.8   | 96.6 | 72.2  | 85.4  |
| NSR+HCPC                             | 57.8  | 91.6   | 94.6 | 83.2  | 84.0  |
| <i>ChartFC (out-of-distribution)</i> |       |        |      |       |       |
| GRPO                                 | 99.0  | 99.0   | 100  | 99.6  | 99.0  |
| GRPO+HCPC                            | 78.0  | 92.4   | 96.6 | 87.2  | 79.0  |
| NSR                                  | 51.4  | 91.0   | 96.2 | 55.6  | 56.0  |
| NSR+HCPC                             | 49.2  | 90.2   | 95.6 | 61.4  | 56.0  |

Table 3: Per-tag emission rate (%) on both benchmarks. On ChartFC, the `<think>` and `<table>` tags collapse hardest under NSR variants (51.4–61.4%, vs. GRPO’s 99.0–99.6%). GRPO+HCPC also degrades OOD but milder ( $\sim 78$ –87%), consistent with the table-dependence finding in §5.2.

## D Pairwise Statistical Comparisons

We computed paired-bootstrap 95% confidence intervals ( $n_{\text{boot}}=10,000$ ) and exact two-sided McNemar  $p$ -values for all 20 method-pair  $\times$  Pass@ $K$   $\times$  benchmark comparisons. Among the trained configurations (GRPO, GRPO+HCPC, NSR, NSR+HCPC), two pairwise gaps reach  $p < 0.05$ : GRPO vs. NSR on ChartFC Pass@4 (+3.8 pp,  $p=0.034$ ; reported in §5.3) and GRPO vs. NSR+HCPC on ChartQA Pass@1 (+5.6 pp,  $p=0.029$ ). All comparisons involving the untuned BASE model are significant on Pass@1 ( $p < 0.05$ ). All other pairs, including the headline GRPO vs. GRPO+HCPC comparison (−0.8 pp on Pass@1,  $p=0.80$ ; +1.2 pp on Pass@4,  $p=0.52$ ), do not reach significance at  $\alpha=0.05$ . The full table is reproducible from the released analysis script (analysis/paper\_stats.py) on the supplied per-sample outputs.

## E HCPC Firing-Rate Distribution

HCPC’s group bonus is zero when fewer than two rollouts pass the correct-path filter ( $|G^+| < 2$ ). The empirical distribution of  $|G^+|$  across the 500 ChartQA test groups for the trained GRPO+HCPC model is:

| $ G^+ $  | 0     | 1     | 2     | 3     | 4     |
|----------|-------|-------|-------|-------|-------|
| fraction | 17.2% | 11.6% | 14.6% | 15.8% | 40.8% |

HCPC fires on  $1 - (17.2 + 11.6) = 71.2\%$  of groups. The 17.2% all-wrong rate is well above the naive Bernoulli bound of 15.7% at  $p=0.62$ , indicating that correctness within a group is pos-

itively correlated, as expected from samples sharing the same chart image.

## F $\Delta_{tbl}$ on the Common-Table Subset

To rule out the confound that  $\Delta_{tbl}$  is computed on different denominators per method, we recompute it on the intersection of samples where both methods produce a parseable table. On the 462 ChartQA samples where both GRPO and GRPO+HCPC produce parseable tables,  $P(\hat{y} = y^* \mid tbl)$  is 0.643 vs. 0.636. The  $\Delta_{tbl}$  gap in Table 1 therefore arises entirely from the  $P(\hat{y} = y^* \mid \text{no tbl})$  arm (0.462 vs. 0.368). We interpret this as evidence that HCPC induces stronger table-dependence, not stronger table-conditioned reasoning accuracy.

## G Per-Answer-Type Breakdown

We split ChartQA test samples by ground-truth answer type and compare GRPO against GRPO+HCPC on Pass@1. Differences are within exact-McNemar noise on numeric ( $n=302$ ,  $-1.7$  pp,  $p=0.63$ ), boolean ( $n=75$ ,  $+4.0$  pp,  $p=0.71$ ), and string ( $n=114$ ,  $-6.1$  pp,  $p=0.30$ ). HCPC does not provide a category-specific accuracy advantage at this scale; the rollout-set robustness gains reported in §5.2 are uniform across answer types rather than concentrated on any single category.

## H Qualitative Example: Format Collapse in Action

Illustration of the NSR collapse and HCPC partial recovery on a single ChartQA sample. The question is “How many categories are there in the chart?” with ground truth  $y^* = 4$ . We show rollout 0 from three methods.

### GRPO (correct, well-formed).

```
<think><type>pie</type><table>{..., "rows":
  ↳ [["Asians", "63"], ...]}</table>
<step-1> The chart is a pie chart...
<step-4> Based on the information provided,
  ↳ there are four distinct categories...
</think>
<answer>4</answer>
```

### NSR (format-collapsed; arrives at right reasoning but wrong answer surface form).

```
<think>First, I'll identify the type of chart
  ↳ and the underlying data table...
<step-1> The chart shows four categories:
  ↳ Asians, Whites, Blacks, and Hispanics. ...
```

```
Final answer:
Four
</think>
<answer>Four</answer>
```

NSR’s rollout contains no `<table>` tag, no `<type>` tag, and emits the reasoning steps inline rather than as structured items. The answer is also expressed in word form (“Four”) rather than the numeric form the eval expects (“4”), so the strict-match metric marks it incorrect despite the reasoning being substantively right.

### NSR+HCPC (table tag recovered; answer still in word form).

```
<think><type>bar</type><table>{"columns":
  ↳ ["Country", "Bar Color"], "rows":
  ↳ [["Asians", "..."], ...]}</table>
<step-1> From the chart...
<step-3> ... there are four distinct
  ↳ categories.
</think>
<answer>Four</answer>
```

HCPC’s group-level bonus on  $C_{table}$  pulls NSR’s policy back toward emitting a `<table>` tag (table fraction 72.2%  $\rightarrow$  83.2% on ChartQA, Table 2). The chart type is incorrectly identified (bar instead of pie), illustrating that HCPC’s partial format recovery is not the same as a full quality recovery, and the answer is still “Four” in word form. This is exactly the dissociation pattern of §5.4: format is partly restored, but the answer-surface alignment that drives strict-match accuracy is governed by a separate channel.